

10(a)(i)	<table><tr><th>crop</th><th>number of stomata</th><th>number of epidermal cells</th><th>stomatal index / %</th></tr><tr><td>cotton</td><td>6</td><td>32 ;</td><td>15.8</td></tr><tr><td>groundnut</td><td>9</td><td>36</td><td>20.0</td></tr><tr><td>mung bean</td><td>5</td><td>12</td><td>29.4</td></tr><tr><td>sorghum</td><td>8</td><td>20</td><td>28.6</td></tr><tr><td>tomato</td><td>7</td><td>25</td><td>21.9 ;</td></tr></table>	crop	number of stomata	number of epidermal cells	stomatal index / %	cotton	6	32 ;	15.8	groundnut	9	36	20.0	mung bean	5	12	29.4	sorghum	8	20	28.6	tomato	7	25	21.9 ;	2
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10(a)(ii)	any two from: 1 allows sufficient, carbon dioxide uptake / gaseous exchange ; 2 reduces, transpiration / water loss ; 3 stomata on lower surface less exposed to, high light intensity / wind ; ora 4 allows more space for photosynthesis ;	2																								

Question	Answer	Marks
10(b)	ignore references to respiration any four from: <i>carbon dioxide concentration is decreasing</i> 1 07:00 to 09:00, light intensity / temperature, is limiting factor ; 2 as, light intensity / temperature, is increasing rate of photosynthesis is increasing ; <i>carbon dioxide concentration is, constant / low</i> 3 09:00 to 17:30 carbon dioxide concentration is limiting factor ; 4 as rate of photosynthesis is high ; <i>carbon dioxide concentration is increasing</i> 5 17:30 to 22:00, light intensity / temperature, is limiting factor ; 6 as, light intensity / temperature, is decreasing the rate of photosynthesis is decreasing ; <i>carbon dioxide concentration is, constant / high</i> 7 22:00–06:00 / 07:00, light intensity is limiting factor ; 8 as, rate of photosynthesis is low / no photosynthesis ;	4
10(c)	1 thylakoid (membrane) ; 2 ATP and reduced NADP used to convert GP to TP ; 3 ATP used in regeneration of RuBP ;	3